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SUBJECT:ETHICAL HACKING

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CO4 -ASS3-CASE STUDY

1. Apache Web Server Security Assessment

Possible Attack Vectors

The attacker may exploit:

- **Outdated Apache server** – old versions may contain known vulnerabilities.
- **Hidden directories** – attackers may find /admin, /backup, or configuration files.
- **SQL Injection** – unsafe user input may affect database queries.
- **XSS** – malicious scripts may be injected into web pages.
- **No WAF** – malicious requests are not filtered before reaching the application.

Kali Linux Methodology

Reconnaissance

`nmap -sV -sC <TARGET_IP>`

This identifies open ports and running services.

`whatweb http://<TARGET_IP>`

This identifies the web technologies used.

Directory Scanning

`gobuster dir -u http://<TARGET_IP> \`
`-w /usr/share/wordlists/dirb/common.txt`

This helps find hidden directories.

Vulnerability Scanning

`nikto -h http://<TARGET_IP>`

Nikto checks the web server for common vulnerabilities.

Prevention

- Regularly update Apache and Linux.
- Use a Web Application Firewall (WAF).
- Validate user input.
- Use HTTPS.
- Monitor server logs.
- Disable unnecessary services.

2. SQL Injection and XSS Attack

SQL Injection

SQL Injection happens when user input is directly used in a database query. An attacker may manipulate the query and access sensitive information.

Testing with Burp Suite

burpsuite

Steps:

1. Open Burp Suite.
2. Intercept the request.
3. Send the request to **Repeater**.
4. Test how the application handles unexpected input.
5. Check for database error messages.

Testing with SQLMap

```
sqlmap -u "http://<TARGET>/page?id=1" --batch
```

This can help identify possible SQL injection vulnerabilities on an authorized target.

XSS

XSS occurs when user input is displayed without proper filtering or encoding.

It may allow malicious scripts to run in another user's browser.

CIA Impact

- **Confidentiality:** Customer data may be stolen.
- **Integrity:** Database records may be changed.
- **Availability:** The application or database may be disrupted.

Prevention

- Use parameterized SQL queries.
- Validate user input.
- Encode output before displaying it.
- Hide detailed error messages.
- Use secure cookies.
- Implement HTTPS and a WAF.

3. Evil Twin Attack

Theory

An **Evil Twin** is a fake Wi-Fi access point that copies the name (SSID) of a legitimate Wi-Fi network.

Example:

Real Wi-Fi: Company_WiFi

Fake Wi-Fi: Company_WiFi

A user may accidentally connect to the fake network. The attacker can then attempt to intercept unprotected traffic or trick users into entering credentials.

Detection Using Kali Linux

Enable monitor mode:

```
sudo airmon-ng start wlan0
```

Scan Wi-Fi networks:

```
sudo airodump-ng wlan0mon
```

Look for:

- Same SSID with different BSSID.
- Unknown access points.
- Suspicious signal strength or channels.

Analyze Using Wireshark

wireshark

Useful filters:

dns

http

Prevention

- Use WPA3 or WPA2-Enterprise.
 - Use 802.1X authentication.
 - Enable Protected Management Frames.
 - Monitor rogue access points.
 - Train users not to connect to suspicious Wi-Fi.
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4. Wireless to Web Application Attack

Attack Chain

Weak Wi-Fi Password



Unauthorized Access



Network Scanning



Find Web Server



Find Vulnerabilities



Access Confidential Data

Step 1: Network Discovery

```
nmap -sn 192.168.1.0/24
```

This identifies active devices on the network.

Step 2: Port Scanning

```
nmap -sV -sC <SERVER_IP>
```

This finds open ports and services.

Step 3: Web Server Scanning

```
nikto -h http://<SERVER_IP>
```

Step 4: Directory Discovery

```
gobuster dir -u http://<SERVER_IP> \
-w /usr/share/wordlists/dirb/common.txt
```

Prevention

Wireless Security

- Use strong passwords.
- Use WPA3 or WPA2-Enterprise.
- Separate employee and guest Wi-Fi.

Network Security

- Use firewall.
- Use VLAN segmentation.

- Implement IDS/IPS.

Web Application Security

- Strong authentication.
 - Secure session management.
 - Input validation.
 - Role-based access control.
 - Use a WAF.
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5. Wi-Fi Deauthentication Attack

Theory

A deauthentication attack is a type of **Denial of Service (DoS)** attack.

The attacker abuses or spoofs Wi-Fi management frames, causing users to disconnect from the access point.

Attacker



Deauthentication Frames



Wi-Fi Users Disconnect



Network Availability Decreases

Detection Using Kali Linux

Enable monitor mode:

```
sudo airmon-ng start wlan0
```

Monitor Wi-Fi traffic:

```
sudo airodump-ng wlan0mon
```

Capture packets:

```
sudo airodump-ng --write capture wlan0mon
```

Analyze using Wireshark:

```
wireshark capture-01.cap
```

Use this filter:

```
wlan.fc.type_subtype == 0x0c
```

This helps identify deauthentication frames.

Impact

- Users lose Wi-Fi connection.
- Online work is interrupted.
- Productivity decreases.
- Network availability is affected.

Prevention

- Enable **Protected Management Frames (802.11w / PMF)**.
- Use WPA3.
- Deploy Wireless IDS/IPS.
- Monitor suspicious Wi-Fi activity.
- Update router and access point firmware.